

Eden Danielsen–Jensen

Ph.D. Candidate in Mathematics

Department of Mathematics, Brigham Young University

edendj@math.byu.edu • eden-dj.github.io/ACADEMIC-WEBSITE

Education

Brigham Young University

Ph.D. in Mathematics

Expected Graduation: August 2027

- Advisor: Jennifer Brooks.
- Dissertation: Group-invariant CR Maps from Spheres to Hyperquadrics (working title).
Investigates the possible target dimensions and signatures of group-invariant CR maps from spheres in $\mathbb{C}^2$ to hyperquadrics, including constructions and regions where no such maps exist.

University of the Sunshine Coast

Bachelor of Science (Honours), Mathematics — First Class Honours

2022

- Advisors: Robert McDougall & Lauren Thornton.
- Honours thesis: *A Revised Introduction for the Base Semisimple Class*.
Used class operators to reinterpret existing results in radical theory concerning the base-semisimple class.

University of the Sunshine Coast

Bachelor of Science — GPA 6.96/7.00

2021

- Major: Mathematics
- Minor: Chemistry

Research Interests

Several Complex Variables, CR Geometry, proper holomorphic mappings, group-invariant CR maps, hyperquadrics, invariant polynomial maps, computational methods in complex analysis.

Publications

Submitted

- Eden Danielsen–Jensen, Dusty Grundmeier, Abdullah Al Helal, Valentin Kunz, Jiří Lebl, Ming Xiao, and Weixia Zhu. *Degree-Three Rational Sphere Maps: Sharp Denominator Region and Gram Normal Forms*. [arXiv:2608.29451](https://arxiv.org/abs/2608.29451).

In Preparation

- Jennifer Brooks, Eden Danielsen–Jensen, Dusty Grundmeier, and Alex Lee. *Lower Bounds for Invariant CR Maps*.

Presentations

Signatures for Group-Invariant Monomial Maps from Spheres to Hyperquadrics

- Joint Mathematics Meetings (JMM), Washington, DC January 2026
- MAA Intermountain Section Meeting, Provo, UT March 2026
- BYU Student Research Conference, Provo, UT February 2026

A Revised Introduction for the Base Semisimple Class

- Australian Mathematical Society Conference, Sydney, NSW December 2022
- Australian Algebra Conference, Sunshine Coast, QLD December 2022

Research Activities

Collaborative Research Workshop in Several Complex Variables

Institute for Advanced Study (IAS), Princeton NJ

June 2026

Teaching Experience

Instructor of Record — Brigham Young University

- | | | |
|-------------------|---------------------------|-------------|
| • MATH 213 | Elementary Linear Algebra | Summer 2026 |
| • MATH 109 | Precalculus | Winter 2026 |
| • MATH 116 | Essentials of Calculus | Fall 2025 |
| • MATH 112 | Calculus I | Spring 2025 |

Teaching Assistant / Recitation Leader — Brigham Young University

- | | | |
|-------------------|----------------------|----------------------|
| • MATH 341 | Theory of Analysis I | Fall 2025, Fall 2026 |
| • MATH 113 | Calculus II | Fall 2026 |

Grader — Brigham Young University

- | | | |
|-------------------|-------------------------------|-------------|
| • MATH 570 | Matrix Analysis | Winter 2025 |
| • MATH 510 | Numerical Linear Algebra | Fall 2025 |
| • MATH 314 | Calculus of Several Variables | Summer 2024 |

Sessional Academic Tutor — University of the Sunshine Coast

- | | | |
|-------------------|-------------------------------------|---------------------|
| • MATH 104 | Introductory Calculus | Semester 2 2023 |
| • MATH 103 | Introduction to Applied Mathematics | Semester 1 2023 |
| • SCI 110 | Science Research Methods | Semester 1 & 2 2023 |

Service & Outreach

- | | |
|---|--------------|
| • Graduate Student Cohort Representative, BYU Mathematics | 2024–Present |
| • Organizer, Graduate Student Writing Group, BYU Mathematics | 2026–Present |
| • Organizer, Graduate Professional Development Workshops, BYU Mathematics | 2025–Present |
| • Graduate School Q&A Panelist, BYU Mathematics | 2025, 2026 |
| • Creator and Facilitator, Mathematical Support Sessions, UniSC | 2023 |
| • Conference Organization Support, Australian Algebra Conference | 2022 |

Honors & Awards

- | | |
|---|-----------|
| • Vice-Chancellor's Honours Scholarship, University of the Sunshine Coast | 2022 |
| • Thompson Excellence Scholarship, University of the Sunshine Coast | 2019–2021 |
| • Rural and Regional Enterprise Scholarship | 2019–2021 |

Professional Memberships

- | | |
|---|--------|
| • American Mathematical Society | Member |
| • Mathematical Association of America | Member |
| • Australian Mathematical Society | Member |
| • Women in Mathematics Special Interest Group (WIMSIG) AustMS | Member |